

Project Proposal Template

The project proposal is how the course will judge the progress of your own project. Illustrated below is a template that you should use in your Week 2 project update blog.

Project Goal:	Develop an interactive ctf styled site to teach about cyber electromagnetic security.
Project Description (more depth):	Level 1: Emergency Tunnel Broadcast A transport authority needs to broadcast emergency information inside a tunnel without requiring drivers to manually change radio stations. Participants investigate how the information is transmitted and how the broadcast system operates. • Provided resources: live spectrum view, audio output, frequency controls, and a Python interface for interacting with the simulated RF environment. • Participant tasks: identify the communication channel, recover the emergency message, and transmit a valid broadcast. • Learning focus: RF communications, signal observation, and protocol discovery. Level 2: Protected Communications After deployment, the authority introduces authentication to prevent unauthorised devices from controlling tunnel infrastructure, including roadside signs and emergency broadcasts. • Provided resources: packet captures, spectrogram views, protocol logs, a Python packet generator, and selected embedded C source code. • Participant tasks: reverse engineer the protocol, understand the authentication mechanism, and generate a valid command. • Learning focus: signalling protocols, authentication mechanisms, and communication security.

Level 3: Radio Jammers Are Nasty

Security auditors suspect that weaknesses in identifiers, keys, or protocol design could allow attackers to manipulate the communication system.

- Provided resources: communication captures, embedded source code fragments, maintenance documentation, and access to a simulated configuration interface.
- Tasks: identify vulnerabilities, recover credentials, and demonstrate unauthorised control of roadside infrastructure or emergency broadcasts.
- Learning focus: replay attacks, protocol exploitation, embedded security, and electronic attack concepts.

Level 4: Adaptive Communications

Following earlier incidents, the authority deploys a simple monitoring and channel-selection system to improve resilience against interference and malicious activity.

- Provided resources: spectrum display, packet monitor, trust score dashboard, event logs, and protocol activity history.
- Tasks: influence system behaviour while avoiding defensive detection mechanisms by sending very slow morse code.
- Learning focus: adaptive communications, intrusion detection, trust modelling, and CEMA-style decision manipulation.

Final Challenge: Capture the Tunnel

Participants combine RF analysis, protocol reverse engineering, authentication exploitation, and detection evasion techniques to compromise the simulated tunnel communication network.

- Provided resources: access to all previous interfaces and system components.
- Participant objectives: transmit an unauthorised emergency broadcast, manipulate roadside infrastructure, and maintain control while staying below defined detection thresholds.
- Success measures: mission completion, communication impact, and detection score.

Project Schedule:	Week 2	- Initial Planning of Design Stages and Project Structure and Data Requirements	
	Week 3	- Technical Planning of level mechanisms - Development of fundamental tunnel system application with no security measures - Development of fundamental flag logic and ctf mechanism - Communications packet interface setup	Note: Project Proposal Due Monday 4PM with Week 2 Portfolio
	Week 4	- Setup of underlying application communication protocol - Build and validate security features such as - o encryption methods - o Keys - o User input mechanism checks etc	
	Week 5	- Setup the user interactive frontend, that has a RF data display - Interactive RF frontend for users to control - Frequency/signalling output - Consoles and packet displays - Frontend and backend connection	
	Week 6	- Culmination of previous elements for final task application creation - Validation of systems and flags	
	Week 7	- Report Documentation Development - Feature Improvement and debugging any issues found - Additional security measures added for increased difficulty	
	Week 8	- Final Edits of Report - Presentation Preparation	Note: Final Report Deliverable Due Friday 4PM 24th July

Project Deliverables/Outcomes:	• Live site/gui that interfaces with a backend application to simulate an RF system • Demonstration of security measures that can be taken and explicit vulnerabilities • Working flagging system, security measures do not let user break systems from previous levels
---------------------------------------	---